Ehsan Sharafian

Ph.D. Candidate, Department of Mechanical Engineering, University of Maine, U.S.

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PROFESSIONAL SUMMARY

Ph.D. candidate seeking a Summer 2027 internship in wearable technology. Experienced in developing integrated wearable systems for gait analysis and rehabilitation using IMU sensing, embedded hardware, smartphone applications, machine learning, and haptic feedback. Published six peer-reviewed research papers, including work in Journal of Biomechanics, IEEE TNSRE, Sensors, and IEEE Access.

EDUCATION

Ph.D. in Mechanical Engineering | University of Maine, U.S. **Sep 2023 - Expected Dec 2027**

GPA: 4.00/4.00 | Dissertation: Smart Real-Time Gait Training System

TECHNICAL SKILLS

Programming & ML: Python, MATLAB, C/C++, Java, NumPy, Scikit-learn, PyTorch, CNNs

Engineering Tools: SolidWorks, Simulink, PLC programming, Android Studio, Git

Wearable Systems: IMUs, sensor fusion, embedded hardware integration, PCB

Motion Analysis: OpenCV, OpenSim, gait analysis, activity recognition, feature extraction, SPSS

RESEARCH & ENGINEERING EXPERIENCE

Graduate Research Assistant | Biorobotics and Biomechanics Lab, University of Maine, U.S. **Sep 2023 - Present**

- Developed a gait monitoring system using IMU, implemented on a smartphone, achieving 96% walking condition classification accuracy and <1.1 cm foot trajectory error in real time.
- Developed an activity recognition classifier for older adults, achieving 99% accuracy for normal locomotion and 93% for atypical stair patterns.
- Built a haptic gait training prototype with embedded hardware and an Android application, improving walking speed by up to 30% with minimal cognitive demand in older adults.
- Integrated OpenCV to track foot motion and analyzed gait mat data to study obstacle crossing mechanisms.

Mechanical and Control Systems Engineer | Sina Robotics and Medical Innovators Co. **Dec 2022 - Sep 2023**

- Developed PLC control logic and validated inverse kinematics in Simulink for a telesurgical laparoscopic robotic platform.
- Built a Python GUI on Raspberry Pi, incorporating feedback from 6 surgeons through direct communication during testing.

Research Assistant | Amirkabir University of Technology **Sep 2018 - Sep 2020**

- Optimized automotive spot-welding robot trajectories using GA and PSO, achieving 24% shorter computed cycle times.
- Developed a screw theory formulation to evaluate joint load distribution in a Delta robot for efficient force transmission.

LEADERSHIP

Graduate Student Senator, Department of Mechanical Engineering, University of Maine **Sep 2025 - Present**

- Represent graduate student concerns in senate meetings, strengthening communication with faculty and university leadership.

SELECTED PUBLICATIONS

- **Ehsan Sharafian**, Babak Hejrati. "Real-Time Foot Height Estimation and Activity Classification Using a Foot-Mounted IMU Implemented on a Smartphone." *Sensors* (2026).
- **Ehsan Sharafian**, Colby Ellis, Babak Hejrati. "Real-Time Activity Recognition Using Minimal Biomechanical Features: A Lightweight IMU-Based Classifier for Older Adults." *IEEE Access* (2025).
- **Ehsan Sharafian**, Colby Ellis, Ben Sidaway, Marie Hayes, Babak Hejrati. "The Effects of Real-Time Haptic Feedback on Gait and Cognitive Load in Older Adults." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2025).